Forecasting Iranian Inflation from Mixed-Frequency Market Signals: An Interpretable Machine-Learning Approach

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***Abstract*—Empirical work on Iranian inflation is limited more by data availability than by method. This paper assembles and validates a monthly panel of observable inflation predictors from reputable sources with programmatic access. The panel joins 12 daily foreign-exchange, gold and coin series (51,792 observations), 20 monthly global commodity series and 252 monthly consumer-price observations, aligned on the Solar Hijri calendar on which Iranian price statistics are defined. Agreement between independent providers (gold in USD, $r = 0.9996$ over 559 months) validates extraction and calendar handling jointly, and the mapped index reproduces officially published annual inflation to within 0.7 pp. Across 56 rolling-origin one-step-ahead forecasts, Ridge regression improves on a random walk by 24% in RMSE; a Diebold–Mariano test does not find this margin significant ($p = 0.101$), and no model survives a Holm correction across the eight compared. In this small sample, penalised linear models outperform both tree ensembles. Twelve features describing the within-month shape of the daily series account for most of the accuracy gain over monthly means. Block-level permutation importance assigns about 60% of predictive content to currency-linked variables, but an out-of-sample ablation shows that the gold and coin block provides no incremental predictive value once the exchange-rate block is included. The importance ranking should therefore not be interpreted as evidence of an independent expectations channel. Annual inflation also shows a marked level shift after 2018 (Chow $F = 29.5$). Overall, the results support a cautious, prediction-focused interpretation of variable attribution in a small, highly collinear macroeconomic panel and highlight the value of time-aware validation and out-of-sample ablation.**

***Index Terms*—inflation forecasting, Iran, exchange-rate pass-through, interpretable machine learning, rolling-origin evaluation, structural breaks, Solar Hijri calendar**

I. Introduction

Iran has experienced sustained high inflation for over a decade, with the consumer price index rising by roughly 88% year-on-year in mid-2026. The data needed to study it is scattered across institutions that publish in different calendars, on different base years, and, in the case of the domestic primary sources, without the programmatic access that an automated and refreshable pipeline requires.

A. Related work

Prior work relevant to this paper falls into three strands. The first concerns the determinants of Iranian inflation. Ture and Khazaei [1] fit a vector error-correction model to quarterly data for 2004–2021 and find that money growth drives inflation in the long run while depreciation and deficits act over both horizons; on the exchange-rate channel, Ebrahimi and Madanizadeh [2] document that pass-through in Iran is not constant, and Abtahi [3] finds it meaningful only in high-inflation regimes. Both pass-through results are testable on our panel, and Section VII recovers the monetary part of [1] on annual data. The second strand is inflation forecasting. Stock and Watson [5] showed that beating a naive benchmark out of sample is difficult even for the United States, and Faust and Wright [10] confirmed that simple benchmarks remain hard to beat across a wide range of models. In the data-rich setting, Medeiros et al. [4] find that machine-learning methods, in particular random forests, improve on standard benchmarks for U.S. inflation, and Coulombe et al. [11] identify nonlinearity and regularisation as the ingredients that matter. Those results are obtained on hundreds of predictors and several hundred months; the small-sample, emerging-market case is far less studied, and for Iran the applied literature is thin [6]. The third strand is interpretable machine learning in economics. Permutation importance and SHAP values [12] are now standard tools for attributing predictive content, but their behaviour under correlated regressors is known to be problematic: Hooker et al. [13] show that permuting one variable while holding its correlates fixed evaluates the model on unrealistic inputs and can misallocate importance, and Strobl et al. [14] document the same issue for tree-based importance. The recommended remedies are conditional or group-wise procedures and out-of-sample refits, which is what the block ablation in Section VI implements.

Against these strands the gap is specific: no study of Iranian inflation (i) builds and validates a predictor panel with within-month information from daily market series, (ii) evaluates it with time-aware nested validation and multiple-comparison correction, and (iii) checks the resulting attribution against an out-of-sample ablation in a highly collinear design. The paper claims no new forecasting method; it evaluates how much can be attributed, and how reliably, from the data that can actually be obtained.

The contribution is fourfold. First, we construct and validate a monthly predictor panel that preserves within-month information from daily market series. Second, we treat calendar alignment as an empirical question, pin it against published official rates, and report the alternative we rejected. Third, the evaluation design uses rolling-origin validation, nested time-aware hyperparameter selection, Diebold–Mariano tests with a small-sample correction and a Holm adjustment across models, together with ablations that examine whether block-level predictive importance persists out of sample. Fourth, we state the limitations explicitly.

II. Data Sources and Access Constraints

Sources differ mainly in how they can be collected. Our aim was a pipeline that fetches every series programmatically and reproducibly, so that the panel can be rebuilt and refreshed with up-to-date data at any time. Judged against that requirement, the domestic primary sources (`amar.org.ir` of the Statistical Centre of Iran, SCI; `cbi.ir` and `tsd.cbi.ir` of the Central Bank of Iran, CBI; and `tsetmc.com`) unfortunately did not serve us well: they offer no documented public API, no stable download endpoints and no machine-readable time-series interface that our tools could rely on for automated retrieval. Rather than build a fragile scraper on top of them, we followed the usual practice in this literature and used reputable international sources with proper programmatic access, namely the World Bank, Yahoo Finance, the Syracuse Iran Data Portal and the Iranian market aggregator TGJU. Because these sources revise or move their files, collection was done in two passes and the raw payloads were retained.

For the dependent variable, the World Bank Global Inflation Database [7] compiles monthly consumer and producer price indices for 209 economies and is retrievable programmatically. For Iran it provides 252 monthly headline observations (1383-01 to 1403-12 Solar Hijri), 168 months of core CPI and 122 of producer prices. Two properties of this series matter throughout: it is a secondary compilation rather than the primary SCI release, and it ends 17 months before the driver series do. Both are discussed as limitations in Section IX.

III. Dataset Construction

A. The calendar grid

Iranian price statistics are defined on Solar Hijri (Jalali) months, which begin on the 19th to 22nd day of a Gregorian month and never on the 1st. Mapping month n to month n therefore introduces an error of about ten days at every observation, which propagates into every lag and growth rate. We use the Solar Hijri month as the grid and aggregate daily and Gregorian-monthly series into it: daily series are reduced to Solar Hijri monthly means, and Gregorian-monthly series are stamped mid-month and assigned to the Solar Hijri month containing that date. We verified that this assignment is lossless: every complete Solar Hijri year in the commodity block holds exactly twelve observations. The price index requires more care, because the source workbook labels Solar Hijri observations with Gregorian dates without documenting the mapping; Section IV-A resolves the offset empirically.

B. Base-year splicing and schema

The Iranian CPI has been rebased three times, from base year 1383 to 1390, 1395 and 1400. Segments are chain-linked on their overlaps, and the implementation raises an exception if consecutive segments do not overlap; on a known series the routine reproduces the original to a maximum absolute error of 1.4×10^{-14}. Every cleaned series carries Gregorian and Jalali dates, an identifier, value, unit and source. Prices are stored in rial rather than toman, no series is transformed destructively, and missing observations appear as explicit nulls on a complete calendar.

Three constructed features are used later. The gold rial premium, the log difference between domestic and global gold prices, captures the wedge between domestic and global gold valuations. The coin-to-gold ratio, the Emami coin price over the 18-carat gram price, is used as a market-based proxy for expectations through the premium paid over bullion content. An explicit Nowruz indicator captures the seasonal concentration of consumption around the Iranian new year, which no Gregorian seasonal dummy reproduces.

C. Keeping the shape of the daily block

Reducing a month of daily quotes to its mean discards the distribution the mean came from: a steady 2% weekly slide and a flat month ending in a 10% drop need not affect next month's prices alike. We therefore retain twelve within-month statistics for the free-market dollar, the Emami coin and the 18-carat gram: the second-half mean relative to the first-half mean, which locates the move within the month; the closing quote relative to the monthly mean; the standard deviation of daily log returns, a realised volatility; and the largest single-day return. Returns are computed within months only, never across a month boundary, and months with fewer than twelve quotes are left missing rather than estimated from a thin sample. Each statistic is complete by the end of the month it describes, so all of them can be lagged as a block with the other drivers.

IV. Validation

Pipeline tests establish that the code does what it claims; they cannot establish that the data is correct. For that we rely on agreement between providers with no common upstream source (Table I).

The first row is the most informative: the TGJU parser and the calendar transformation are both ours, the World Bank series passes through neither, and agreement at $r = 0.9996$ over 559 months would not arise if either were wrong. Unit consistency is confirmed physically: an Emami coin contains

TABLE I
INDEPENDENT CROSS-SOURCE AGREEMENT.

Comparison	n	r	Median gap
Gold USD: TGJU vs. WB Pink Sheet	559	0.9996	0.75%
Brent: Yahoo vs. Pink Sheet	228	0.9922	1.93%
Annual infl.: WB vs. Iran Data Portal	55	0.974	1.40 pp

gold equivalent to 9.76 grams of 18-carat gold, so the coin-to-gold ratio has a floor; the observed ratio (9.75 to 12.68) sits on it within a dealer spread, whereas a rial–toman confusion would displace it tenfold. Calendar handling converts known dates (Nowruz 1400, the leap-year Esfand of 1399) exactly, and the index contains no month-on-month step below -0.6%.

A. Pinning the index to the Solar Hijri calendar

Two independent arguments fix the offset at zero. Structurally, the Iran series begins in exactly 2004-04 and ends in exactly 2025-03, Iranian rather than Gregorian year boundaries, as expected if Solar Hijri data are stamped with the Gregorian month in which each period begins. Empirically, annual inflation derived from the mapped monthly index reproduces the SCI's published rates: 47.1% derived against 46.5% official for 1401, 41.4% against 40.7% for 1402, and 32.4% against 32.5% for 1403, all within 0.7 pp. We also record one alternative that was rejected. Shifting the stamp two months forward improves agreement with the World Bank's own annual series (r from 0.969 to 0.992), but that series itself departs from the official figures by about 3 pp in several years, so the improvement reflects the idiosyncrasies of a secondary compilation rather than a corrected calendar. Officially published rates are the appropriate reference; a second compilation only corroborates.

V. TWO REGIMES OF CHRONIC INFLATION

The monthly panel sits inside a longer annual record (Fig. 1). From 1973 to 2017 Iranian inflation was chronic but bounded: it averaged 18.4% over 45 years, was in double digits in 93% of years, and never approached hyperinflation. Four decades of double-digit inflation are consistent with a persistent inflation regime rather than a sequence of isolated shocks. After 2018 the level shifts: the post-2018 mean is 36.8% over 8 years, and no year falls below 18%, a floor above most peaks of the earlier regime. A mean-shift Chow test at 2018 gives $F = 29.5$ ($p = 1.5 \times 10^{-6}$), and scanning candidate break years from 1990 to 2022 places the strongest break at the same event.

On monthly data the more recent candidate break is not identifiable. Monthly depreciation averaged 2.55% (s.d. 6.77) before 1403 against 4.02% (s.d. 5.95) afterwards, yet a Chow test at 1403-01 does not reject stability in the depreciation mean ($F = 1.19$, $p = 0.28$) or in the pass-through regression ($F = 0.87$, $p = 0.42$) on 29 post-break months. A sup-F scan over the middle 70% of the sample places the strongest break at 1397-04 (2018); its distribution is non-standard when the

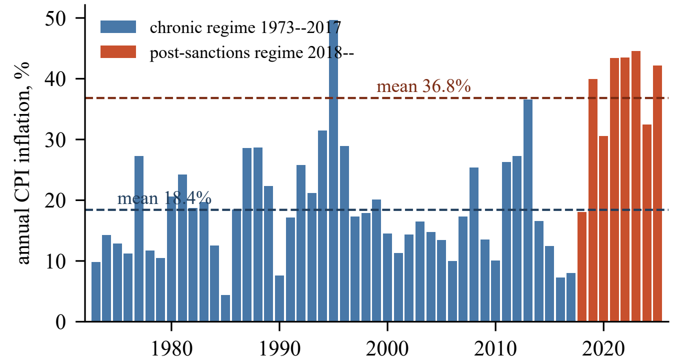

Fig. 1. Annual CPI inflation, 1973–2025, with regime means (dashed): 18.4% for the chronic regime, 36.8% from 2018 onward.

date is chosen from the data [8], so we report the location without a p-value.

VI. MODEL AND RESULTS

A. Specification

The target is the month-on-month log change of the headline index, in per cent. We avoid a year-on-year target because its overlapping windows induce serially correlated forecast errors by construction and inflate the apparent accuracy of every model. Predictors fall into four blocks. Exchange-rate movements enter at one-, three- and twelve-month horizons together with a six-month volatility term, and the monthly dollar change is split into its positive and negative parts because prices ratchet: a rial that loses 10% and recovers it does not leave the price level where it started. Gold and coin variables include the premium and the coin-to-gold ratio. Imported commodities include Brent, food, energy and the global gold price in US dollars; the last is classified as an imported commodity rather than an expectations proxy, and classifying it by its `gold_` prefix instead would move about 6 pp of attribution between blocks. Autoregressive terms use lags 1, 2, 3, 6 and 12 and moving averages. The twelve within-month statistics and seasonal terms complete the set. All drivers are lagged one month as a block, so no future information enters the design. No imputation is used at any stage: features are constructed on the full calendar, then a month enters the design only if the target and all 51 features are observed for it (listwise deletion), which leaves 128 months (1393-05 to 1403-12). The 51 features are fixed in advance by construction rules, not selected on the data; the larger candidate count cited in Section IX refers to a different, audit-stage column set. This is a predictor-rich small-sample setting in which regularisation is appropriate.

Evaluation is rolling-origin with an expanding window: we fit on the first 72 months, predict the next, advance one month and refit, which yields 56 genuine one-step-ahead forecasts (1399-05 to 1403-12). Random k-fold cross-validation would train on the future to predict the past and is not used. Within each training window the penalised

TABLE II
OUT-OF-SAMPLE ACCURACY, 56 ONE-STEP-AHEAD FORECASTS. RATIOS BELOW ONE BEAT THE RANDOM WALK (RW); DIR. IS DIRECTION HIT RATE; $p(H)$ IS HOLM-ADJUSTED.

Model	RMSE	RMSE/RW	Dir.	DM p	$p(H)$
Ridge	1.225	0.762	68%	0.101	0.705
Combo(ML)	1.247	0.776	64%	0.118	0.706
ElasticNet	1.248	0.776	70%	0.122	0.706
Combo(Ridge+RW)	1.294	0.805	68%	0.051	0.409
RandomForest	1.328	0.826	61%	0.229	0.916
GradBoost	1.455	0.905	64%	0.509	1.000
AR	1.486	0.924	66%	0.490	1.000
RandomWalk	1.608	1.000	—	—	—
Mean	1.772	1.102	66%	0.496	1.000

models are pipelines of a `StandardScaler` and the estimator, so scaling and all hyperparameters are fitted on that window's rows only. Ridge selects its penalty from 40 log-spaced values in $[10^{-3}, 10^{3}]$ and ElasticNet its penalty and l_1 ratio (in $\{0.1, 0.5, 0.7, 0.9, 0.95, 1\}$) by inner five-split `TimeSeriesSplit` on squared error; time-ordered inner folds cost about one percentage point of accuracy relative to random ones. The tree models use fixed settings (random forest: 500 trees, minimum leaf 3, 40% of features per split; gradient boosting: 400 iterations, learning rate 0.05, 15 leaves, minimum leaf 8, ℓ_2 penalty 1.0) and a fixed seed. The benchmarks are a random walk, the training-window mean and a ridge autoregression on inflation lags 1, 2, 3 and 12; the learners are Ridge, ElasticNet, the random forest and gradient boosting, plus two equal-weight combinations fixed before scoring: Combo(ML), the mean of the four learners, and Combo(Ridge+RW), the mean of Ridge and the random walk. Accuracy differences use the Diebold–Mariano statistic [9] under squared-error loss with the Harvey–Leybourne–Newbold correction [15], with Holm-adjusted p-values across the eight comparisons.

B. Accuracy

Ridge is the most accurate specification, with an RMSE of 1.225 against 1.608 for the random walk, an improvement of 24%, and it calls the direction of change correctly in 68% of months (Table II, Fig. 2). The margin is not statistically significant ($\mathrm{DM} = 1.67$, $p = 0.101$). The lowest unadjusted p-value in the table belongs not to the most accurate model but to the ridge–random-walk combination ($p = 0.051$): combination raises RMSE slightly while making the loss differential more consistent from month to month, which is the usual argument for forecast combination. After the Holm adjustment, no model reaches conventional significance. On this evidence, a claim that machine learning beats the naive forecast for Iranian inflation would go beyond what 56 observations can support.

The ranking is as informative as the level. Penalised linear models outperform both tree ensembles, and gradient boosting falls behind the simple autoregression, a pattern consistent with the flexible learners overfitting at this sample size; the gain over the random walk is captured by the penalised linear

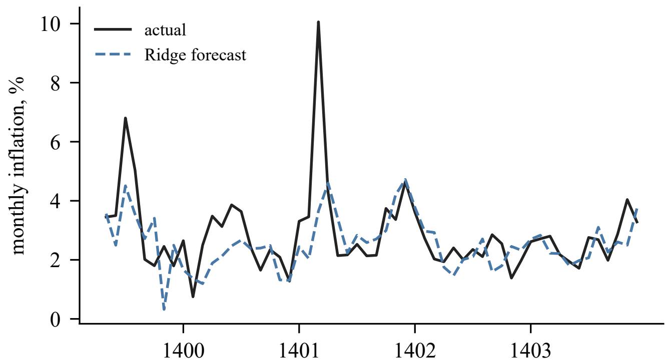

Fig. 2. Out-of-sample monthly inflation: actual vs. Ridge, 56 one-step-ahead forecasts, 1399-05 to 1403-12. The model tracks level and turning points but understates the sharpest spikes.

specifications. Given that Iranian applications have tended toward neural architectures [6], this result is relevant. The gain that does exist comes largely from the daily series: without the twelve within-month statistics the RMSE ratio is 0.82 rather than 0.76, roughly the whole margin over a specification built from monthly means, and four of the ten highest-ranked features are within-month statistics, with the largest single-day dollar move ranked second overall.

C. Attribution and its out-of-sample check

Attribution is in-sample: the Ridge pipeline is refitted on all 128 rows, permutation importance is computed per feature on those rows (30 permutations, squared-error scoring), and block shares are the within-block sums as a percentage of the total after clipping negatives at zero. Mean |SHAP| uses a linear explainer on the same fitted model and standardised rows. On this basis the exchange-rate block carries 36.7% and the gold and coin proxies 22.8%, so the two currency-linked blocks together account for about 60%, against 25.3% for inflation persistence, 10.0% for imported commodities and 5.2% for seasonality. Separately, the feature-level ranking based on mean absolute SHAP values places the gold rial premium first (mean |SHAP| 0.093) and the largest single-day dollar move second. These are predictive attribution measures, not causal effects: a variable can improve forecasting by anticipating a common shock without transmitting one, and the currency blocks are highly collinear, so the split between them is less reliable than their combined contribution.

To test these cautions, we withhold each block in turn and re-run the 56 forecasts (Table III). Removing the within-month statistics raises RMSE by 8.0% and removing the exchange-rate block by 5.1%, whereas removing the gold and coin block lowers RMSE by 1.2%: the model is marginally better without it. The two results are not contradictory; they answer different predictive questions. In a collinear design, permutation importance can distribute shared signal across correlated variables, while an ablation asks whether a block contributes incremental out-of-sample information conditional on the remaining blocks. Here the gold and coin variables do

TABLE III
ABLATION: 56 FORECASTS RE-RUN WITH ONE BLOCK WITHHELD. POSITIVE ΔRMSE MEANS ACCURACY DETERIORATES WITHOUT THE BLOCK.

Block withheld	RMSE/RW	ΔRMSE
none (full model, 51 features)	0.762	—
within-month shape	0.823	$+8.0\%$
inflation persistence	0.825	$+8.3\%$
exchange rate	0.801	$+5.1\%$
seasonality	0.779	$+2.2\%$
gold and coin	0.753	-1.2%
imported commodities	0.762	-0.1%

TABLE IV
INTEGRATION ORDER, ESTIMATION SAMPLE ($n = 128$). ADF NULLS A UNIT ROOT; KPSS NULLS STATIONARITY.

Series	ADF p	KPSS p	Verdict
Monthly inflation (target)	0.000	0.010	ambiguous
usd_d1	0.000	0.100	stationary
usd_d12	0.262	0.100	ambiguous
gold_rial_premium	0.971	0.010	unit root
coin_gold_ratio	0.393	0.010	unit root
food_d12	0.158	0.094	ambiguous
usd_vol6	0.031	0.080	stationary

not provide such incremental information once the exchange-rate block is present. The combined currency-linked importance should therefore not be interpreted as evidence for two independent economic channels. We recommend reporting ablation evidence alongside attribution rankings in collinear economic data.

D. Stationarity and robustness

Table IV reports the stationarity diagnostics. The ADF test takes a unit root as its null and the KPSS test takes stationarity as its null, so agreement between them is more informative than either test alone. The monthly dollar change is classified as stationary, whereas the two tests give conflicting evidence for the monthly inflation target. The gold rial premium and the coin-to-gold ratio are classified as unit-root processes and enter the design in levels. Since the premium is the highest-ranked feature, this matters. Refitting with the two unit-root levels dropped (their first differences remain in the design) moves the RMSE ratio only from 0.76 to 0.77 and the currency-linked share from 63% to 56%, consistent with the ablation finding that these variables are close to redundant.

Two further checks were run. For calendar robustness, refitting with the price index shifted one month in either direction leaves the zero-offset specification the most accurate (RMSE ratio 0.76 against 0.92 and 0.78), which corroborates Section IV-A from a different direction; the currency-linked share ranges from 53% to 63% across the three, so the qualitative conclusion does not depend on the offset even though the exact figure does. For pass-through stability, the coefficient of monthly inflation on lagged monthly depreciation over a rolling 48-month window runs from $+0.078$ at the start of the sample to $+0.096$ at the end, within a range of $[-0.101, +0.192]$. It is imprecisely estimated in any single window but shows no trend: on this sample, pass-through drifts rather than breaks, which is consistent with regime dependence [3], [2] while remaining stable over the estimable years.

E. Conditional projection for the period the index does not cover

The driver series extend 17 months beyond the last published index observation. Feeding the observed drivers into the fitted model, with the autoregressive terms carried forward recursively, gives an ex-post conditional projection for 1404-01 through 1405-05; it is not a real-time forecast that would have been available at the end of 1403. The projected month-on-month log changes are cumulated, $\exp(\sum_h \hat{\pi}_h/100) - 1$, giving 67% over the 17-month horizon; annualising that cumulative figure to a 12-month equivalent gives about 44%, and the cumulative change over the first twelve projected months alone is also 44%. Resampling the 56 out-of-sample residuals and passing them through the same recursion 2,000 times gives a 95% interval of 40% to 67% annualised. A year-on-year (point-to-point) headline rate near 88% has been reported for mid-1405; that figure is quoted from public reporting and has not been reconciled here against the SCI release, and it is a point-to-point rate whereas the projection is a cumulative path, so the two must be verified on a common definition, source and horizon before the gap is treated as a formal forecast error. Until that check is completed, the comparison is best regarded as a diagnostic indication that the post-1403 episode may contain information not represented in the fitted panel; it does not by itself identify which omitted mechanism is responsible.

VII. EXPLORATORY MONETARY EVIDENCE AT ANNUAL FREQUENCY

The monthly forecasting panel does not contain monetary aggregates or fiscal-financing variables, so it cannot identify a structural monetary mechanism. We therefore treat the available annual series only as supplementary descriptive evidence. At annual frequency, broad-money growth correlates with inflation at $+0.41$ when lagged one year, compared with $+0.21$ contemporaneously, while lagged net domestic-credit growth correlates at $+0.38$. These correlations are consistent with earlier evidence on the role of monetary expansion [1], but they are not causal estimates and they come from a different frequency and sample coverage than the monthly forecasting exercise. Direct fiscal measures are even less informative because the available series end before the episodes that dominate the recent sample. The practical implication is therefore modest: monthly monetary variables would be a valuable addition to future versions of the predictor panel and may reduce the extent to which market-price variables proxy omitted macroeconomic conditions.

VIII. DISCUSSION

The main findings should be interpreted as predictive rather than structural. The exchange-rate block provides incremental out-of-sample information: removing it raises RMSE by 5.1%. By contrast, the gold and coin block receives substantial in-sample attribution but does not improve out-of-sample accuracy once exchange-rate variables are present. This distinction illustrates why importance rankings alone are insufficient in a collinear design; block ablations provide a useful complementary check on whether a proposed predictor group adds information conditional on the others.

A second implication concerns data access. The primary monthly CPI and monetary series are not available through stable programmatic interfaces; making them available in machine-readable form would permit direct robustness checks against the primary target and allow monetary variables to enter the monthly design. Conclusions here are therefore restricted to predictive performance and incremental information in the observed panel, not to policy or causal prescriptions.

IX. Limitations

(i) The index ends in 1403-12 while the drivers run to 1405-05, so the recent period can only be examined through a conditional ex-post projection and cannot be scored against the target series used for estimation. (ii) The target is a secondary compilation. It validates well (Section IV-A), but a headline result should rest on the primary SCI figures, and we name the variable `cpi_headline_wb` throughout so that it cannot be confused with them. We did not interpolate annual data into a monthly proxy, since that would make monthly attribution circular. (iii) The sample is small relative to the number of predictors. At the data-audit stage the panel offers 141 months in which the target and the core drivers are jointly present against 54 engineered columns, of which only 33 months carry every column; the model does not use that column set. The design of Section VI is built from 51 fixed features with listwise deletion and no imputation, giving 128 months (about 2.5 per feature). (iv) The attribution shares are predictive, not causal, and permutation importance can split shared signal arbitrarily between correlated regressors, which is why the ablation is reported alongside. (v) The autoregressive terms assume that last month's index is available, which the publication lag prevents; the loss is confined to the persistence block, and the design is pseudo-out-of-sample rather than an operational forecast. (vi) The IMF has conducted no Article IV consultation with Iran since 2018, and the SCI and the central bank publish divergent CPI figures; free-market exchange, gold and coin prices, being market-generated, provide an external check on official price data.

X. Conclusion

The data situation is better than is generally assumed: 51,792 daily observations across 12 market series, validated against independent providers to within 1%, and 252 monthly index observations whose calendar alignment is pinned against official rates. On that basis, the best model reduces RMSE relative to a random walk by 24%, although the difference is not statistically significant over the 56 forecast months. Penalised linear models outperform the nonlinear ensembles in this sample, a pattern consistent with the limited sample size. The ablation further narrows the predictive interpretation: removing the exchange-rate block worsens out-of-sample accuracy, whereas removing the gold and coin block does not. Thus, the importance ranking should not be read as evidence of an independent expectations channel.

The 17 months beyond the target series can be examined only through an ex-post conditional projection, which falls well below the rate reported for mid-1405; the two must be harmonised to one definition and horizon before the gap is read as a forecast error. More generally, attribution in a small, collinear macroeconomic panel should be paired with time-aware validation and out-of-sample ablation, and the absence of programmatically accessible monthly monetary and primary CPI series remains the central limitation. All results are computed from a frozen, checksummed data snapshot, and every source used is publicly accessible through the interfaces described in Section II, so the panel can be rebuilt independently from the provenance given there; the authors' code and snapshot are not distributed with the paper.

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